THE BERNSTEIN PROBLEM

WED APRIL 9, 2025
MAT 355

OUTLINE

0)-State the problem + a few pre-reqs + MS examples i) Minimal surfaces ii) Harmonics + Liouville type theorem
iii) Analytic proof of Bernstein's Problem in $\mathbb{R}^3$ iv) Cones + Geometric proof in $\mathbb{R}^3 (\Rightarrow \mathbb{R}^4)$

0)-OVERVIEW

THE PROBLEM

Bernstein's Problem in the form of a conjecture:
If the graph of a real-valued function on $\mathbb{R}^{n-1}$ is an entire minimal surface in $\mathbb{R}^n$ then the function is linear ($\Rightarrow$ graph is a (hyper)plane)

True for $n \leq 8$, false for $n \geq 9$

PRE-REQS

$u: \mathbb{R}^n \to \mathbb{R} \Rightarrow$ graph of $u = \Sigma = \{(x, u(x)) : x \in \mathbb{R}^n\} \subset \mathbb{R}^{n+1}$ i.e. $z = u(x,y)$ for $\mathbb{R}^2$
In order for a surface to be a graph, every point in $\mathbb{R}^n$ must correspond to precisely one value
function is linear $\Rightarrow u(x) = a_1 x_1 + \dots + a_n x_n + b$ for $x \in \mathbb{R}^n$
Entire minimal surface $\Rightarrow$ globally minimizes area $\Leftrightarrow$ satisfies the MSE $\Leftrightarrow H = 0$
i.e. $\Sigma = \{(x, u(x)) : x \in \mathbb{R}^n\} \subset \mathbb{R}^{n+1}$ is an entire minimal surface if:

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\left(\frac{\partial_i u}{\sqrt{1+|\nabla u|^2}}\right) = 0 \Leftrightarrow H = 0$$

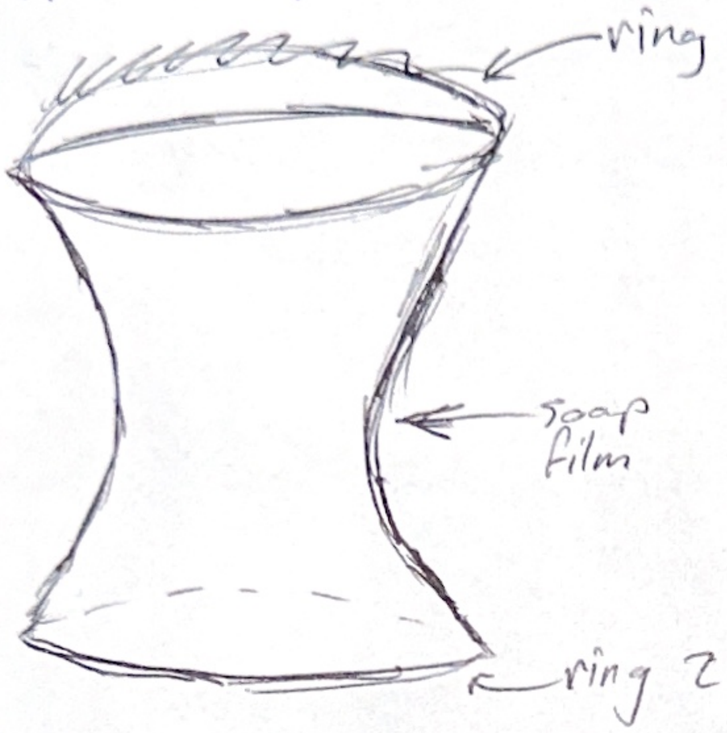
← MENTION THAT I WRITE ZEROS LIKE THIS

Recalling $\operatorname{div} F \equiv \nabla \cdot F$ where $\nabla = (\partial_{x_1}, \partial_{x_2}, \dots, \partial_{x_n}) \in \mathbb{R}^n$

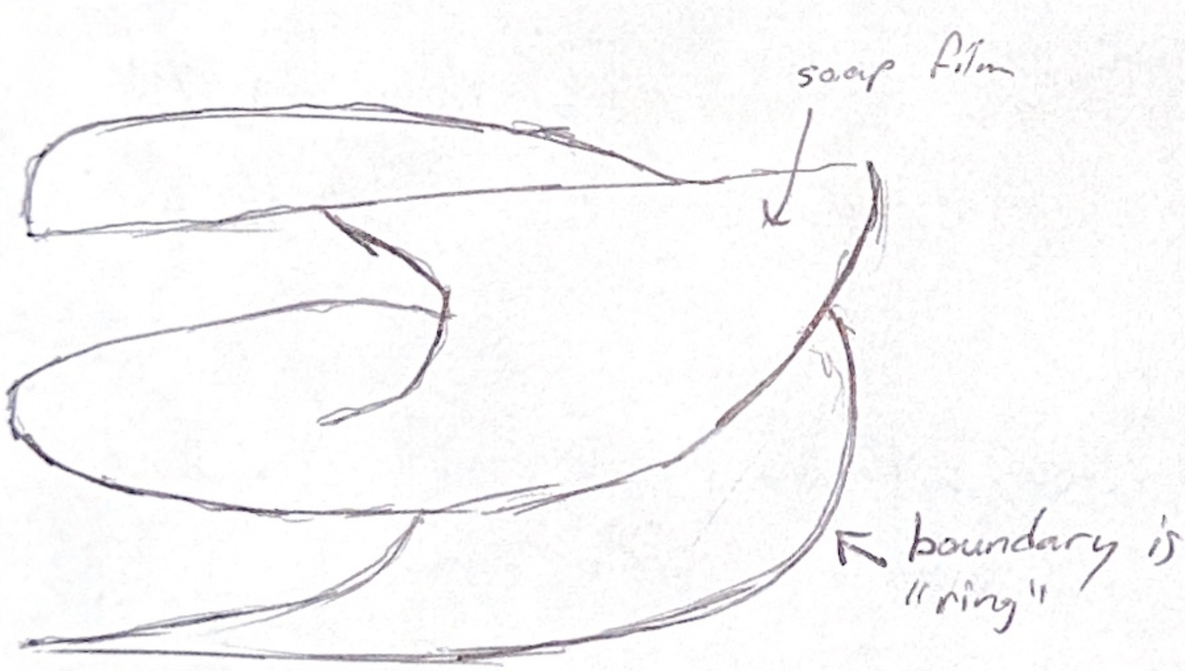
SEE BACK FOR EXAMPLES OF MINIMAL SURFACES

EXAMPLES

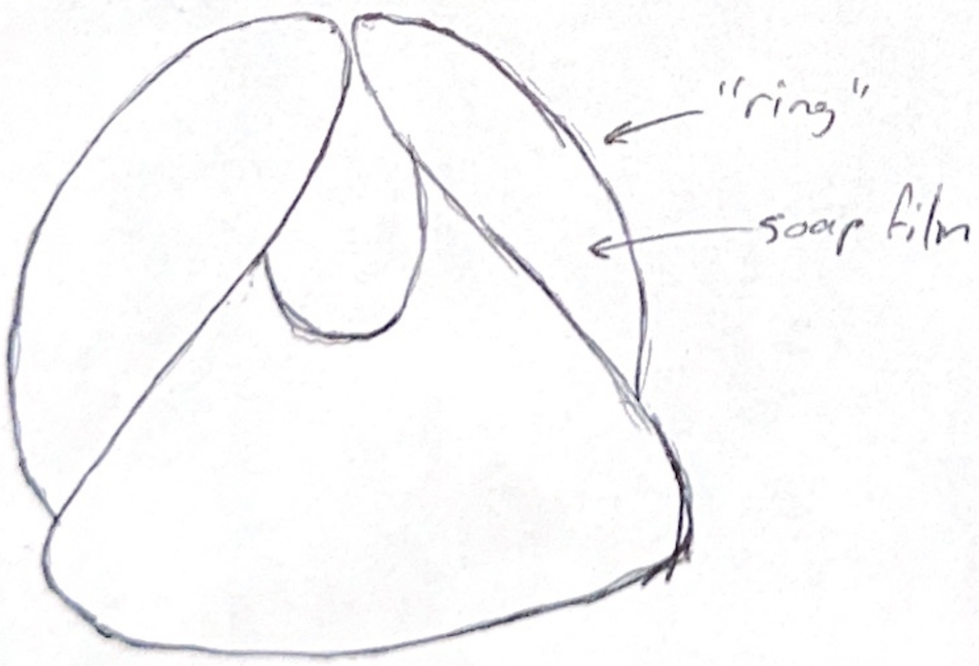
CATENOID:



HELICOID



ENNEPER SURFACE



i)-Minimal Surfaces

History + Defs:

In the 18th and 19th centuries, with the advent of the calculus of variations, the study of surfaces that minimize their area for a given boundary—minimal surfaces—began

This provided a way to find functions that extremize a "functional" (e.g. area)

Functional—maps function to a scalar, $F: \mathcal{A} \to \mathbb{R}$, $F[u] = \int_D \mathcal{L}(x, u(x), \nabla u(x))\,dx$ ($\mathcal{A}$: some space of fnctns e.g. C^∞)

essentially $\mathcal{L}$ is what we want to minimize (i.e. want $u = \arg\min_v F[v]$)

using this, Lagrange was able to derive the MSE (see below)

In 1776, Jean Baptiste Meusnier showed that surfaces with zero mean curvature are minimal

He also discovered that the helicoid and catenoid are (locally) minimal

In the 19th century many advances were made:

- Joseph Plateau, studying soapfilms, posed the question: for a given boundary can we find a surface bounded by this curve with minimal area? Answer is yes. Solving the problem just amounts to solving the MSE with the B.C. $u = \emptyset$ on the given boundary.
- Many more minimal surfaces were discovered,
- Connections to complex analysis (Weierstrass-Enneper parametrization)

In the early 20th century (1915), Bernstein proved his theorem (for an entire minimal graph ($\in \mathbb{R}^3$ over $\mathbb{R}^2$) in $\mathbb{R}^3$

In the 60s many more advancements were made in this problem:

- 1962 (Fleming)—using area-minimizing surfaces that are cone-like at the origin (i.e. minimizes area near cone's apex) he showed that $\exists$ non-planar area-minimizing cones in $\mathbb{R}^3$, providing a new proof of Bernstein's theorem.
- 1965—De Giorgi showed that if $\exists$ non-planar area-minimizing cones in $\mathbb{R}^{n-1}$ then the analogue of Bernstein's thm holds in $\mathbb{R}^n$, proving $\mathbb{R}^4$ (MAYBE NOT, ONLY WIKIPEDIA)
- 1966—Almgren $\exists$ these cones in $\mathbb{R}^4$, extending Bernstein's thm to $\mathbb{R}^5$
- 1968—Simons showed that $\exists$ these cones in $\mathbb{R}^7$, extending it to $\mathbb{R}^8$ (Founder of Renaissance Medallion fund; quant) proposed the "Simons cone" in $\mathbb{R}^8$, showed that it was locally stable and asked whether or not it was globally area-minimizing
- 1969—Bombieri, De Giorgi, and Giusti showed that the Simons cone is indeed globally area minimizing, proving that Bernstein's thm breaks down in $\mathbb{R}^9$, and therefore that nonplanar entirely mimimal graphs exist in $\mathbb{R}^n$ for $n \geq 9$

DERIVATIONS

AREA FUNCTIONAL:

Here we will show it for a surface $\Sigma \subset \mathbb{R}^3$ but the idea can trivially be extended.

define $\Sigma := \{(x,y,u(x,y)) : (x,y) \in D \subset \mathbb{R}^2\}$

Now let's choose the parametrization of this surface: $r(x,y) = (x,y,u(x,y))$

$\Rightarrow r_x = (1,0,u_x) \quad r_y = (0,1,u_y) \Rightarrow |r_x \wedge r_y| = \sqrt{1+u_x^2+u_y^2} = \sqrt{1+|\nabla u|^2}$

Recall (do Carmo): $A(R) = \int_Q |r_x \wedge r_y| dx dy$ where $Q = r^{-1}(R \subset \Sigma) = \{(x,y) \in D : r(x,y) \in R\}$

This gives the area for a (variable) region of a graph produced by a fixed fnctn (u)

instead we want to find the area of a fixed region and allow the fnctn to change

$$\boxed{A[u] = \int_D \sqrt{1+|\nabla u|^2}\, dx dy} \Rightarrow \mathcal{I}(x,y,\nabla u) = \sqrt{1+|\nabla u|^2}$$

Now we want to find u that minimizes this functional (Euler-Lagrange) ← calculus of variations

Euler-Lagrange: $\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{I}}{\partial \dot{x}_i}\right) - \frac{\partial \mathcal{I}}{\partial x_i} = 0$ where $\mathcal{I}(t, x_i(t), \dot{x}_i(t))$ and E-L finds minimizing $x_i(t)$

in our functional: $t \longleftrightarrow x,y \quad x_i(t) \longleftrightarrow u \quad \dot{x}_i(t) \longleftrightarrow \nabla u$

$$\Rightarrow \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{I}}{\partial r_x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial \mathcal{I}}{\partial r_y}\right) = 0 \leftarrow \text{From E-L}$$

quick calculation shows: $\frac{\partial \mathcal{I}}{\partial r_x} = \frac{u_x}{\sqrt{1+|\nabla u|^2}}$ and $\frac{\partial \mathcal{I}}{\partial r_y} = \frac{u_y}{\sqrt{1+|\nabla u|^2}}$

therefore E-L becomes $\frac{\partial}{\partial x}\left(\frac{u_x}{\sqrt{1+|\nabla u|^2}}\right) + \frac{\partial}{\partial y}\left(\frac{u_y}{\sqrt{1+|\nabla u|^2}}\right) = 0$ ← this is the MSE!

$= (u_x, u_y)$

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = \frac{\partial}{\partial x}\left(\frac{u_x}{\sqrt{1+|\nabla u|^2}}\right) + \frac{\partial}{\partial y}\left(\frac{u_y}{\sqrt{1+|\nabla u|^2}}\right) = 0 \quad (*)$$

Thus we have derived the $\mathbb{R}^2$ MSE, could trivially be extended.

CONNECTION TO H:

Recall $H = \frac{1}{2}\frac{eG - 2fF + Eg}{EG - F^2}$, by plugging in the definitions we see:

$$H = \frac{(1+u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1+u_x^2)u_{yy}}{2(1+|\nabla u|^2)^{3/2}} \quad \text{where the numerator} = (*)$$

Therefore, any solution to the MSE satisfies $\boxed{H = 0}$ ← entire graph

ii) HARMONICS

PRE-REQS:

Recall the Laplacian in $\mathbb{R}^n$: $\Delta := \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$

$v: \Omega \subset \mathbb{R}^n \to \mathbb{R}$ is "harmonic" if $\Delta v = 0 \iff v(x_0) = \frac{1}{|B_r(x_0)|}\int_{B_r(x_0)} v(x)\,dx$ ← Mean Value Property (MVP)

"subharmonic" if $\Delta v \geq 0 \iff v(x_0) \leq \frac{1}{|B_r(x_0)|}\int_{B_r(x_0)} v(x)\,dx$

$B_r(x) := $ open ball of radius r centered at $x = \{x \in \mathbb{R}^n : |x| < r\}$ $\quad B_r := B_r(0)$

$|B_r(x)| :=$ volume of $B_r(x)$ $\quad \overline{\Omega} = \Omega \cup \partial\Omega$

$C \subset\subset \Omega$ is "closed" in Ω if it contains all its limit points

$\iff$ if $\{x_n\} \subset C$ converges to $x \in \Omega$ then $x \in C$

$O \subset \Omega$ is "open" if $\forall x \in O \ \exists B_r(x) \subset \Omega$ s.t. $B_r(x) \subset O$ "clopen" $\iff$ closed & open

unique continuation property (UCP) — if $v: \Omega \to \mathbb{R}$ harmonic, C^2 is constant on any subset of Ω then v must be constant throughout Ω ↖ see ⊖ for a proof

PROOFS:

WEAK MAXIMUM PRINCIPLE (WMP)

claim: if $v: \Omega \to \mathbb{R}$ harmonic, $C^2(\Omega)$, $C^0(\overline{\Omega})$ ("continuously extended to its boundary"), bounded on $\Omega \subset \mathbb{R}^n$, then

$$\sup_{x \in \overline{\Omega}} v(x) = \sup_{x \in \partial\Omega} v(x)$$

proof: Assume that v is not constant

then $\exists\, x_0 \in \overline{\Omega}$ s.t. $v(x_0) = \sup_{x \in \overline{\Omega}} v(x)$ ← x_0 is a global max

if $x_0 \in \partial\Omega$ then we are done

suppose $x_0 \in \Omega$:

By MVP: $v(x_0) = \frac{1}{|B_r(x_0)|}\int_{B_r(x_0)} v(x)\,dx$

Since x_0 is a global max $v(x) \leq v(x_0)$ $\forall x \in B_r(x_0)$ $\forall r > 0$

(✗) $\Rightarrow \frac{1}{|B_r(x_0)|}\int_{B_r(x_0)} v(x)\,dx \leq v(x_0)\frac{1}{|B_r(x_0)|}\int_{B_r(x_0)} dx = v(x_0) \Rightarrow v(x) = v(x_0)$ $\forall x \in B_r(x_0)$ ⊕ (Follows from $v(x) \leq v(x_0)$ and ints are equal)

Extending this to Ω by connectedness: ← AT THIS POINT JUST APPLY UCP, STILL NEED ⊖ FOR SMP

$S := \{x \in \Omega : v(x) = v(x_0)\} \neq \emptyset$ ← since $x_0 \in \Omega$

We will now show that S is clopen:

open — $\forall x \in S$, $v(x) = v(x_0)$ thus we can apply MVP as done in ⊕ giving

⊖ $v(x) = v(y)$ $\forall y \in B_r(x)$ thus $B_r(x) \subset S$ ✓

CALL THIS ARGUMENT (*)

CALL THIS ARGUMENT (−)

closed – let $\{x_n\} \subset S$ converge to $x \in \Omega$ then (b/c $v \in C^2(\Omega)$) $\lim_{n\to\infty} v(x_n) = v(x)$

but $\lim_{n\to\infty} v(x_n) = v(x_0)$ since $\{x_n\} \subset S \Rightarrow v(x_0) = v(x) \Rightarrow v(x) \in S$ ✓

It is a standard result from set theory that the only nonempty clopen set on any set is the set itself, thus $S = \Omega$

$\Rightarrow v(x) = v(x_0) = \text{const.}\ \forall\, x \in \Omega$

$\Rightarrow v(x) = \text{const.}\ \forall\, x \in \overline{\Omega}$ ← $\lim_{\substack{x\to x_1 \\ x\in\Omega}} v(x) = \text{const.} = v(x_1)$ ← by continuity for $x_1 \in \partial\Omega$

Thus we have shown that either v attains a max on the boundary of Ω or v is constant on $\overline{\Omega}$ $\Rightarrow$ $\boxed{\sup_{\overline{\Omega}} v = \sup_{\partial\Omega} v}$

STRONG MAXIMUM PRINCIPLE (SMP)

claim: if $v : \Omega \to \mathbb{R}$ subharmonic, C^2 on $\Omega \subset \mathbb{R}^n$ connected, open, v attains a local max on Ω (i.e. $\exists\, x_0 \in \Omega$ s.t. $v(x) \le v(x_0)\ \forall x \in B_r(x_0)$) then v is constant on $\overline{\Omega}$.

proof: Note that this claim is almost identical to what we showed in (*) of the proof for WMP. The key difference is that here v is subharmonic, not harmonic. Therefore to apply WMP we would like to show that v is harmonic (globally):

Since x_0 is a local max, observe the following:

all partials critical pts vanish $\to Dv(x_0) = 0$ $\quad \left.\frac{\partial^2 v}{\partial x^{i2}}\right|_{x_0} \le 0\ \forall i \Rightarrow \Delta v(x_0) = \sum_{i=1}^{n} \frac{\partial^2 v}{\partial x^{i2}} \le 0$ ← i.e. Hessian $(D^2 v(x_0))$ is negative semi-definite

$\Rightarrow \Delta v(x_0) \le 0$ but $\Delta v \ge 0\ \forall x$ by subharmonic property ↙ since $\Delta v(x_0) = 0 \Rightarrow$ locally flat

$\Rightarrow \Delta v(x_0) = 0 \Rightarrow \Delta v(x) = 0\ \forall x$ in a small neighborhood of x_0 (i.e. $\forall x \in B_r(x_0)$) ← small r

By MVP: $v(x_0) \le \frac{1}{|B_r(x_0)|}\int_{B_r(x_0)} v(x)\,dx \le v(x_0)\frac{1}{|B_r(x_0)|}\int_{B_r(x_0)} dx = v(x_0)$ ← since $v(x_0)$ is a local max

$\Rightarrow v(x) = v(x_0)\ \forall x \in B_r(x_0)$

Now follow the same argument as in (−), giving $v(x) = v(x_0)\ \forall x \in \overline{\Omega}$

Moreover, this gives us $\Delta v(x) = 0\ \forall x \in \overline{\Omega}$

Now, applying UCP (or (−)), we see: $\boxed{v(x) = \text{const.}}\ \forall x \in \overline{\Omega}$ □

EXTENSION OF SMP (ESMP)

claim: if $v : \mathbb{R}^n \to \mathbb{R}$ subharmonic, C^2, $v(x) \le M\ \forall x \in \mathbb{R}^n$, $\lim_{|x|\to\infty} v(x) = M$, then $v(x) = M\ \forall x \in \mathbb{R}^n$

i.e. v cannot approach M at infinity w/o attaining it somewhere

proof: proceed by contradiction, i.e. assume $v(x) \ne M\ \forall x \in \mathbb{R}^n$

Because $v(x) \to M$ as $|x| \to \infty$, it follows that $\forall\, \varepsilon > 0\ \exists\, R > 0$ s.t.

$v(x) \ge M - \varepsilon\ \forall\ |x| > R$

Moreover, $\forall x_0$ satisfying $|x_0| > R+1$, $v(x_0) \le \frac{1}{|B_1(x_0)|}\int_{B_1(x_0)} v(x)\,dx \le M - \varepsilon$ ← since $v(x) \le M-\varepsilon\ \forall x \in B_1(x_0)$

bounded ← Not really, I just want to use WMP □

But since $|x_0| > R+1 > R$, $v(x_0) \geq M-\varepsilon \Rightarrow v(x_0) = M-\varepsilon$

Similarly for $0 < \varepsilon' < \varepsilon$ $\exists\, R' > R$ s.t. $v(x) = M-\varepsilon'$ $\forall\, |x| > R_2 > R_1$

$\Rightarrow$ fix $|x_1| > R' \Rightarrow v(x_1) = M-\varepsilon' = M-\varepsilon$ impossible ($\varepsilon_1' = \varepsilon$ ruled out by construction)

in fact, the only ε that works $\forall r$ is $\varepsilon = 0 \Rightarrow v(x) = M \Rightarrow$ a contradiction □

LIOUVILLE THEOREM FOR SUBHARMONICS (LTS)

claim: $v: \mathbb{R}^n \to \mathbb{R}$ subharmonic, C^2, bounded above, then v is constant

(if $\Delta v \geq 0$)

proof: proceed by contradiction $\Rightarrow \exists\, x_0 \in \mathbb{R}^n$ s.t. $v(x_0) < \sup_{x \in \mathbb{R}^n} v(x) =: M$

define the following $\varepsilon := M - v(x_0) > 0$ $\quad M_R := \sup_{\overline{B}_R} v = \sup_{\partial B_R} v$

clearly: $M_R \leq M$, $R \leq R' \Rightarrow M_R \leq M_{R'}$, $\lim_{R\to\infty} M_R = M$

case 1: M_R attained in B_R

By SMP, $v = $ const. on B_R ~~$\Rightarrow v(x_0) = M_R$ $\Rightarrow v(x_0) + \varepsilon_2 \Rightarrow$ contradiction~~

$\Rightarrow v = $ const. on $\mathbb{R}^n$ (by $\ominus$) $\Rightarrow$ contradiction ↯

case 2: M_R is attained on ∂B_R

$\Rightarrow v(x) < M$ $\forall x \in \mathbb{R}^n$ ($\Rightarrow v = $ const. by ESMP?)

$\forall\, R_1 < R_2$: $M_{R_1} \leq M_{R_2} \leq M$ and $M_R \to M$ as $R \to \infty$

$\forall\, \varepsilon > 0$ $\exists\, R_\varepsilon$ s.t. $M_{R_\varepsilon} > M - \varepsilon$

$\Rightarrow v(x) \leq M_{R_\varepsilon}$ $\forall\, x \in \overline{B_{R_\varepsilon}}$ ← By WMP

$\Rightarrow M_{R_\varepsilon} \to M$ as $\varepsilon \to 0 \Rightarrow v(x) \leq M - \varepsilon$ for arbitrarily large R_ε

$\Rightarrow v(x) \to M$ as $|x| \to \infty$ but never attains it

$\Rightarrow v(x) = M$ $\forall x$, thus a contradiction has been reached □

A bit of intuition for SMP: $\Delta u \geq 0 \Rightarrow$ "upward curving"

$\Rightarrow$ local max iff flat

iii) - BERNSTEIN'S PROBLEM IN $\mathbb{R}^3$

Let $u: \mathbb{R}^2 \to \mathbb{R}$ be a solution to the MSE

Let $\sum := \{(x,y,u(x,y)) : (x,y) \in \mathbb{R}^2\} \subset \mathbb{R}^3$ be its graph

define the following auxillary function: $v := \sqrt{1+|\nabla u|^2}$

LEMMA 1 - v is subharmonic

proof: This proof is nearly all computation in order to show $\Delta v \geq 0$

recall $\Delta v = v_{xx} + v_{yy}$, thus the calculation is straightforward:

$$v_x = \frac{u_{xx}u_x - u_y u_{xy}}{v}$$

$$v_{xx} = \frac{(u_{xx}^2 + u_x u_{xxx} + u_{xy}^2 + u_y u_{xxy})v^2 - (u_{xx}u_x + u_y u_{xy})^2}{v^3}$$

There are analagous results for v_y and v_{yy}, giving us:

$$\Delta v = v_{xx} + v_{yy} = \frac{1}{v^3}\left(v^2(u_{xx}^2 + u_{yy}^2 + 2u_{xy}^2) + (\ldots) - (u_x u_{xx} + u_y u_{xy})^2 - (u_x u_{xy} + u_y u_{yy})^2\right)$$

(third derivative terms, all cancel)

$$\geq \frac{1}{v^3}\left((u_{xx}^2 + u_{yy}^2 + 2u_{xy}^2) - |\nabla u|^2(u_{xx}^2 + 2u_{xy}^2 + u_{yy}^2)\right) \leftarrow \text{by Cauchy-Schwarz}$$

$$(u_x u_{xy} + u_y u_{yy})^2 \Downarrow (u_x^2 + u_y^2)(u_{xy}^2 + u_{yy}^2)$$

$$= \frac{(u_{xx}^2 + u_{yy}^2 + 2u_{xy}^2)(v^2 - |\nabla u|^2)}{v^3} \leftarrow = 1$$

$$= \frac{u_{xx}^2 + u_{yy}^2 + 2u_{xy}^2}{v^3} \geq 0 \Rightarrow \Delta v \geq 0 \qquad \square$$

LEMMA 2 - v is bounded above

proof: to prove this it suffices to show that ∇u is bounded above

it is a standard procedure in solving PDEs to place estimates on solutions

in particular, people have derived the following a priori estimate for the gradient of solutions to the MSE:

$$\sup_{\mathbb{R}^n} |\nabla u| \leq C_1 \exp(C_2(n)A) \text{ where } n \text{ is the dimension}$$

intuitively, we see that the normal is given by:

$$N = \frac{(-u_x, -u_y, 1)}{\sqrt{1+|\nabla u|^2}} \Rightarrow N_3 = \frac{1}{\sqrt{1+|\nabla u|^2}}$$

For a graph, Gauss map cannot pass the hemisphere

if $|\nabla u| \to \infty$ then $N_3 \to 0 \Rightarrow$ the Gauss map approaches the hemisphere

$\Rightarrow$ tangent plane is vertical, clearly not "area-minimizing"

So, by Liouville's theorem for subharmonics, we may conclude that v is constant.

$\Rightarrow$ the MSE becomes $\operatorname{div}(\nabla u) = \Delta u = 0 \Rightarrow u$ harmonic

~~$\Rightarrow$ By stuff, we may therefore conclude that ∇u is constant proving Bernstein's problem in $\mathbb{R}^3$~~

But as we don't know that u is bounded so we may not invoke Liouville.

we do know, however $|u_{x_i}(x)| \leq |\nabla u(x)| = C$ ← $|\nabla u| = C$ by Liouville for subharmonics, know from the properties of v

Since derivatives commute with the Laplacian: $\Delta(u_{x_i}) = \frac{\partial}{\partial x_i}(\Delta u) = 0$

$\Rightarrow u_{x_i}$ is harmonic $\Rightarrow u_{x_i} = \text{const.}$

doing this for each i we see $\boxed{\nabla u = \text{const.}} \Rightarrow u$ affine. proving Bernstein's thm on $\mathbb{R}^3$.

IR³-CONES

write $\Sigma := \{(x, u(x)) : x \in \mathbb{R}^2\}$ for brevity

PROPOSITION 1: THE ASYMPTOTIC BEHAVIOR OF AN ENTIRE MINIMAL SURFACE ON $\mathbb{R}^3$ IS THAT OF A CONE

proof: call this minimal surface $\Sigma := \{(x, u(x)) : x \in \mathbb{R}^2\}$ where $u: \mathbb{R}^2 \to \mathbb{R}$ solves the MSE

we see that $u_R(x) := \frac{1}{R} u(Rx)$ also solves the MSE ← LEMMA 1, simple change of variables

This gives a new, scaled minimal surface: $\Sigma_R := \{(x, u_R(x)) : x \in \mathbb{R}^2\}$

By taking $R \to \infty$ we are able to look at the asymptotic behavior of Σ, i.e. even if Σ is locally not conic, these variations go away

LEMMA 2 → Assume that as $R \to \infty$ (really should say a suitable subsequence $R_k \to \infty$, will just write R for brevity) $u_R \to u_\infty$ $\forall x$ ("locally uniformly"?)

define $u_{\lambda R}(x) := \frac{1}{\lambda R} u(\lambda R x) \Rightarrow u_R(\lambda x) = \frac{1}{R} u(R\lambda x)$

$\forall \lambda > 0$ $\Rightarrow u_{\lambda R}(x) = \frac{1}{\lambda}\left(\frac{1}{R} u(\lambda R x)\right) = \frac{1}{\lambda} u_R(\lambda x)$

Observing that $\lambda R \to \infty$ when $R \to \infty$, we see:

$$u_{\lambda R}(x) = \frac{1}{\lambda} u_R(\lambda x) \Rightarrow \lim_{R\to\infty} u_{\lambda R}(x) = \lim_{R\to\infty} \frac{1}{\lambda} u_R(\lambda x) \Rightarrow \lambda u_\infty(x) = u_\infty(\lambda x)$$

Giving us the asymptotic behavoir of the graph:

$$\Sigma_\infty = \{(x, u_\infty(x)) : x \in \mathbb{R}^2\}$$

$\forall x \in \mathbb{R}^2$, $\forall \lambda > 0$, we see $\lambda(x, u_\infty(x)) = (\lambda x, u_\infty(\lambda x))$

{I THINK THE WRITTEN STATEMENT IS INCORRECT, STRICTLY SPEAKING. NOT VERY RIGOROUS, JUST TRY TO EXPLAIN INTUITIVELY ↙}

The only graph that has this invariance for dilations about the origin is the cone.

$\Rightarrow \Sigma_\infty$ is a cone $\Rightarrow \Sigma$ is asymptotically a cone □

LEMMA 1: IF $u: \mathbb{R}^2 \to \mathbb{R}$ SOLVES THE MSE THEN SO DOES $u_R := \frac{1}{R} u(Rx)$ $\forall x \in \mathbb{R}^2$

proof: $\frac{\partial}{\partial x^i} u_R(x) = \frac{\partial}{\partial x_i} \frac{1}{R} u(Rx) = \frac{1}{R} R \frac{\partial}{\partial y^i} u(y) = \frac{\partial}{\partial y^i} u(y) \Rightarrow \nabla u = \nabla u_R$

$y = Rx \Rightarrow \frac{\partial}{\partial x^i} = \frac{\partial y}{\partial x^i} \frac{\partial}{\partial y} = R \frac{\partial}{\partial y^i}$ $\Rightarrow |\nabla u| = |\nabla u_R|$

$\Rightarrow u_R$ solves the MSE. □

Before moving onto lemma 2, we will need the following:

equicontinuous – A family of functions $\{f_n\}: \mathbb{R}^2 \to \mathbb{R}$ is "=" if $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. ←

$|x - y| < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$ $\forall n$

Arzelà-Ascoli theorem: if a family of functions $\{f_n\}$ is equicontinuous and uniformly bounded on a compact set, then $\exists$ a convergent subsequence.

LEMMA 2: THE SEQUENCE $\{R\} \to \infty$ HAS A SUBSEQUENCE $\{R_j\} \to_{\{R\}} \infty$ s.t. $u_{R_j} \to u_\infty$

($\{R\}$ starts at 1 (i.e. $\{R\}_{R \geq 1}$))

proof: our goal is to show that $\{u_{R_j}\}$ (i) equicontinuous and (ii) uniformly bounded on a compact set, so that we may invoke Arzelà-Ascoli to show $u_{R_j} \to u_\infty$ (iii).

(i) fix $K \subset \mathbb{R}^2$ compact s.t. $\forall x \in K$, $|\nabla u_R(x)| \leq C_k$ $\forall R > 0$ ← BY LEMMA 1 AND BOUND ON ∇u

$\Rightarrow$ by MVT, $\forall x, y \in K$, $|u_R(x) - u_R(y)| \leq C_k |x-y|$

$\Rightarrow \forall x, y \in K$ satisfying $|x-y| < \delta = \frac{\varepsilon}{C_k}$ we see $|u_R(x) - u_R(y)| \leq C_k |x-y| < C_k \frac{\varepsilon}{C_k} < \varepsilon$ ✓

So we have shown $\{u_R\}$ to be equicontinuous

(ii) set $\varepsilon = 1$:

(observe. $\forall R$ observe $u_R(\emptyset) = \frac{1}{R} u(\emptyset)$, set $u(\emptyset) := M \Rightarrow u_R(\emptyset) = \frac{M}{R}$

$|u_R(x) - u_R(\emptyset)| < 1 \Rightarrow |u_R(x)| \leq |u_R(\emptyset)| + 1 = \frac{M}{R} + 1$ $\forall |x| < \delta = \frac{1}{C_k}$

$\Rightarrow |u_R(x)| < L := M + 2$ places a uniform bound on u_R $\forall x \in B_\delta$

because K is compact, we can cover it with finitely many of these δ balls (B_δ). At each of them we will have a uniform bound on u_R taking the max of all these bounds then gives us a uniform bound on u_R.

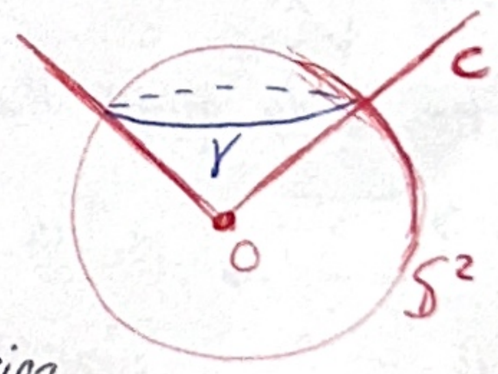
(iii) Therefore, by Arzelà-Ascoli we have shown that $\exists$ a subsequence of $\{u_R\}$ (call it $\{u_{R_j}\}$) s.t. $u_{R_j} \to u_\infty$. □

PROPOSITION 2: THE ONLY AREA-MINIMIZING CONES IN $\mathbb{R}^3$ ARE PLANES.

proof: let C be this area-minimizing cone.

consider $\gamma = C \cap S^2$

clearly, $\text{Area}(C \cap B_R) \propto R^2 \text{Length}(\gamma)$ (*) const. is $\frac{1}{2}$ ← see discussion below



When we applied E-L, we were really just guaranteeing that $\forall$ minimal surface with a small perturbation, ε, the variation (∧ to the of the area) to the first order is 0, i.e. $\frac{d}{d\varepsilon} \text{Area}(C_\varepsilon)\big|_{\varepsilon=0} = 0$ where $C_\varepsilon = \{(x, u_\varepsilon(x)) : x \in \mathbb{R}^2\}$

$u_\varepsilon = u(x) + \varepsilon \underline{v(x)}$ smooth, ε small

observing $\text{Area}(C_\varepsilon \cap B_R) \propto R^2 \text{length}(\gamma_\varepsilon)$ where $\gamma_\varepsilon = C_\varepsilon \cap S^2$

we see: $\frac{d}{d\varepsilon} \text{length}(\gamma_\varepsilon)\big|_{\varepsilon=0} = 0$

$\Rightarrow$ the first variation of the length of γ_ε is $0 \Leftrightarrow \gamma$ is stationary for the length functional

It is a standard result that any closed, ~~stationary~~ smooth curve on a Riemannian manifold (e.g. S^2) is stationary w.r.t. the length functional is a geodesic

Recall that the geodesics of ~~great circles~~ spheres are the great circles

~~is centered at the origin~~

From the geometric picture, it is clear that the only way that γ is a great circle is if C degenerates into the plane.
Hence we have shown that the only area-minimizing cones in $\mathbb{R}^3$ are planes. □

(*): This was actually an implementation of the "coarea formula": ($\int d\mathcal{H}^1$ = length, $\int d\mathcal{H}^2$ = Area...)

" " says: $f:\mathbb{R}^n \to \mathbb{R}$, Lipshcitz: $\int_{\mathbb{R}^n} g(x)\,|\nabla f(x)|\,dx = \int_{\mathbb{R}} \left(\int_{f^{-1}(t)} g(x)\, d\mathcal{H}^{n-1}(x) \right) dt$ ($\mathcal{H}$ = Hausdorff measure)

Lipshcitz: $|f(x)-f(y)| \le C|x-y|$ ← showed our function satisfies this

This formula is essentially Fubini's theorem for level sets, i.e. instead of slicing space into rectangles, we slice it into level surfaces of f.

Easy example: $\int_V \phi(x)\,dV = \int_0^\infty \left(\int_{S_r} \phi(x)\,dS \right) dr$ (S_r is sphere of radius r)

By proving these two propositions, we have provided another proof of Bernstein's theorem in $\mathbb{R}^3$. (i.e. the only entirely minimal surfaces on $\mathbb{R}^3$ are planes) ↑ MAYBE NOT
Moreover, this proves Bernstein's problem in $\mathbb{R}^4$ (De Giorgi 1965)
As we move to higher dimensions, this proof fails because ~~[illegible]~~
in higher dimensions the cross section of the cone with S^{n-1} is much more complicated than just the great circle
Can end up with non-totally geodesic minimal hypersurfaces
Also, the gradient estimates begin to fail.

COUNTEREXAMPLE – SIMONS CONE

The Simons cone, ~~[illegible]~~ $C := \{(x,y) \in \mathbb{R}^4 \times \mathbb{R}^4 : |x| = |y|\} \subset \mathbb{R}^8$ (← actually just $m \ge 4$)

This, written out, is $x_1^2+x_2^2+x_3^2+x_4^2 = x_5^2+x_6^2+x_7^2+x_8^2$

clearly this eq. is quadratic $\Rightarrow C$ is not affine.

C is smooth everywhere other than the origin where it has a singular pt. (H not dfnd)

in fact $H=0$ everywhere other than the origin.

~~to show that it is globally area-minimizing, compare it with other hypersurfaces that coincide with the cone outside a compact set K and then we note that the cone minimizes the area in K~~

This ~~cone~~ in particular was investigated after Simons showed it to be stable at the singular point. (i.e. the second variation of the area along smooth deformations is nonnegative)
$\Rightarrow$ perturbations do not increase the area
Still doesn't prove it though.

~~To prove it, Bombieri De Giorgi Giusti compared it with other hypersurfaces that coincided w/ the cone outside~~

The proof by Bombieri, De Giorgi, and Giusti in 1969 relied on a ~~Fett~~ "foliation" by smooth surfaces: Constructed a family of smooth hypersurfaces asymptotic to ~~the~~ C, foliating $\mathbb{R}^8 \setminus C$. These surfaces act as barriers ensuring that competitor surfaces with smaller areas cannot exist. By analyzing the functional perturbations ε and leveraging the foliation structure they showed that any deviation from the cone increases the area.

~~Note hyperplanes~~

Note: hyperplanes in $\mathbb{R}^n$ for $n \geq 9$ are still entirely minimal

OPEN PROBLEMS

Spherical Bernstein: let $\iota: S^n \longrightarrow S^{n+1}$ be an embedding of an n-sphere s.t. $\iota(S^n)$ is minimal.

Is $\iota(S^n)$ an equator of S^{n+1}

Almgren + Calabi in 1966/7 - true for $n=2$

Hsaing in 1983 - false for $n=4$ and 3-13 odd

Tomter in 1987 - False for odd n

Open for even $n \geq 8$!

Two-valued Bernstein: let $u: \mathbb{R}^n \longrightarrow \{\mathbb{R}, \mathbb{R}\}$ be a two-valued minimal graph is Σ_u the union of 2 entire minimal surfaces in $\mathbb{R}^n$

Rosales in 2016 - true for $n=2$

Hiesmayr in 2020 - true for $n=3$

Open for $n \geq 4$!